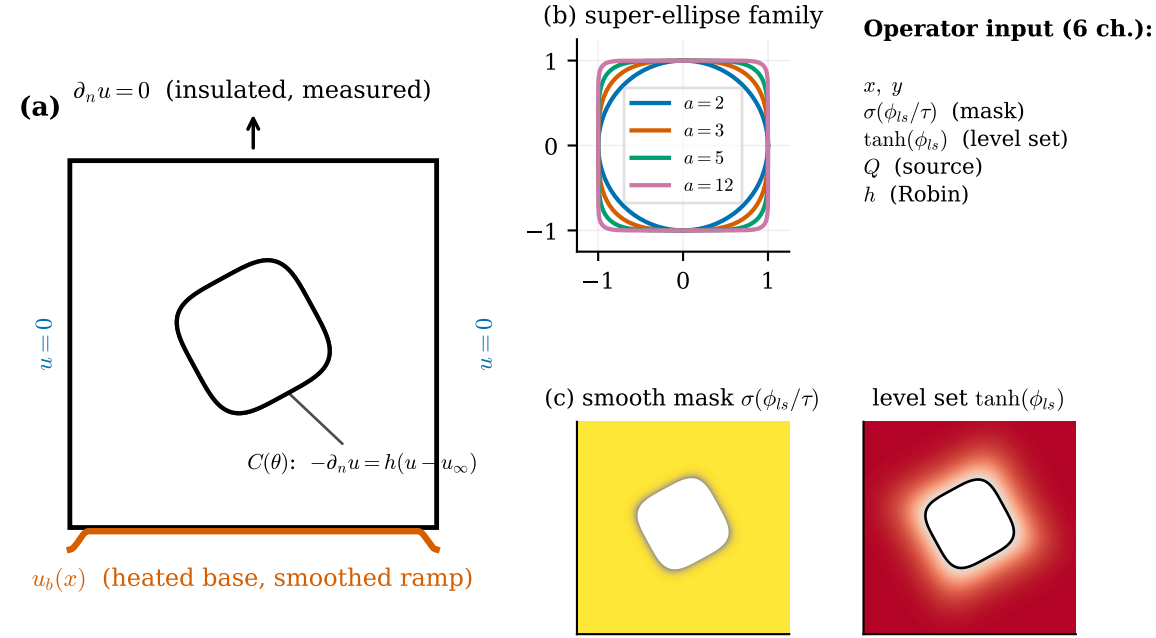

(a)
$\partial_n u = 0$ (insulated, measured)
$u = 0$
$u = 0$
$C(\theta)$: $-\partial_n u = h(u - u_\infty)$
$u_b(x)$ (heated base, smoothed ramp)
(b) super-ellipse family
$a = 2$
$a = 3$
$a = 5$
$a = 12$
1
0
−1
−1
0
1
Operator input (6 ch.):
$x,\ y$
$\sigma(\phi_{ls}/\tau)$ (mask)
$\tanh(\phi_{ls})$ (level set)
Q (source)
h (Robin)
(c) smooth mask $\sigma(\phi_{ls}/\tau)$
level set $\tanh(\phi_{ls})$